

Attachment: Supporting Measurements for Paper #229

All perplexity values are WikiText-2 unless noted. Tables A4-A5 use measured GPU non-FFN timing combined with TLM/CIM FFN service ratios as described in the main response.

Table A1. Qwen Scale, High-Quality WikiText-2 PPL

Model	FP16	MXFP8	TSS 30 dB	WANDA 2:4	Act. 2:4
Qwen3-0.6B	17.3443	17.2754	17.3212 (+0.0458)	40.3042	37.5716
Qwen3-1.7B	14.6644	14.6248	14.4661 (-0.1587)	27.2134	20.9451
Qwen3-4B	11.2364	11.2757	11.2926 (+0.0169)	18.7089	16.5446
Qwen3-8B	8.2722	8.2964	8.3177 (+0.0213)	12.4731	10.6433

Table A2. Llama-3.2-3B Cross-Family PPL

Model	MXFP8	TSS 17 dB	TSS 30 dB	WANDA 2:4	Act. 2:4
Llama-3.2-3B	6.5173	6.7859	6.5271	15.1917	11.0518

Table A3. Qwen Lower-Work Operating Point, 17 dB

Model	MXFP8	TSS 17 dB	WANDA 2:4	Act. 2:4
Qwen3-0.6B	17.2754	19.4307	40.3042	37.5716
Qwen3-1.7B	14.6248	14.9315	27.2134	20.9451
Qwen3-4B	11.2757	13.7060	18.7089	16.5446
Qwen3-8B	8.2964	11.5926	12.4731	10.6433

Table A4. Full-GPU-Relative Hybrid E2E Latency

Measured GPU full/non-FFN times are combined with TLM/CIM FFN service ratios; each gain uses the measured full-GPU latency as the denominator.

Scenario	Method	GPU full ms	GPU non-FFN ms	GPU FFN ms	FFN share	TLM FFN ratio	Hybrid E2E ms	Norm. vs GPU	Gain
decode_s2048	GPU full baseline	22.799	20.710	2.088	9.2%	1.000000	22.799	1.000000	0.000%
decode_s2048	TSS 17 dB	22.799	20.710	2.088	9.2%	0.535680	21.829	0.957471	4.253%
decode_s2048	Mask-CIM 2:4	22.799	20.710	2.088	9.2%	0.875000	22.538	0.988551	1.145%
prefill_s512	GPU full baseline	24.904	20.606	4.299	17.3%	1.000000	24.904	1.000000	0.000%
prefill_s512	TSS 17 dB	24.904	20.606	4.299	17.3%	0.535680	22.908	0.919857	8.014%
prefill_s512	Mask-CIM 2:4	24.904	20.606	4.299	17.3%	0.875000	24.367	0.978425	2.158%

Table A5. Qwen3-1.7B Decode, Common-Interface Comparison

Arbitrary flexible-mask hardware is excluded unless a matched CIM routing/buffering/utilization model is available; Table A5b states the modeled 2:4 Mask-CIM assumptions.

Method	PPL	Norm. lat.	Lat. gain	Norm. throughput	Norm. energy	Energy red.	Norm. area	Basis
Dense MXFP8	14.6248	1.000	0.0%	1.000	1.000	0.0%	1.000	measured full-GPU denominator; dense simulator baseline
TSS 17 dB	14.9315	0.957	4.3%	1.044	0.884	11.6%	1.011	measured GPU split plus TLM FFN ratio; RTL timing/area; proxy dynamic energy
Mask-CIM 2:4	20.9451	0.989	1.1%	1.012	0.925	7.5%	1.114	formula-backed mask ledger; dense reads/payload kept; no packed-sparse issue

Table A5b. Mask-CIM 2:4 Modeling Assumptions

Component	Modeled treatment
Activation read	dense
Mask generation/control	counted
Weight read	selected 2-of-4 entries benefit
Local reduction tree	selected fan-in benefit
Fixed pipeline stages	dense
Output payload	dense
Utilization model	dense-lane issue; no packed sparse issue
Routing/buffering for arbitrary masks	not assigned without matched CIM design
Energy status	formula-backed mask ledger; TSS controller energy remains proxy where labeled
Latency ledger	dense fixed/access/payload/control terms; selected-read and local-tree terms receive 2-of-4 benefit
Energy ledger	activation read + compare + mask payload/control + selected weight/compute/tree + dense output payload

Table A6. Qwen3-1.7B K Sensitivity

A6 is a within-sweep ablation on the same fixed evaluation subset for all K rows; compare absolute PPL within A6, while A1 remains the full-test headline.

K	MXFP8 PPL	TSS 17 dB PPL	Digit-work proxy	Metadata entries vs K=32
8	13.6369	13.5499 (-0.0870)	0.465	4.00x
16	13.5906	13.5840 (-0.0066)	0.470	2.00x
32	13.5924	13.7575 (+0.1651)	0.473	1.00x
64	13.6242	13.7191 (+0.0949)	0.474	0.50x
128	13.5624	13.4459 (-0.1165)	0.474	0.25x

Table A7. Calibration Runtime Examples

Model/configuration	Runtime
Qwen3-1.7B, K=64, 17 dB	2802.9 s
Qwen3-1.7B, K=16, 17 dB	2963.4 s
Llama-3.2-3B, 17 dB	5584.0 s
Llama-3.2-3B, 20 dB	5692.0 s
Llama-3.2-3B, 30 dB	5868.7 s

Table A8. Control-Path Accounting

Control work is counted once and staged one block ahead. The table separates the counted control operations from the timing and storage anchors.

Category	Item	Accounting	Value / policy
Control work	Scale prepass	counted	computes scale offsets and block delay
Control work	Scheduler/window build	counted	consumes block delay, horizon, and input-delay terms
Control work	Input-delay decode/fanout	counted	included in controller path
Control work	Configuration switch	counted	active/shadow buffering
Timing anchor	Controller service	counted	3 staged control cycles
Timing anchor	Data-plane slot	reference	16 cycles/block
Timing anchor	Slack after fill	derived	about 13 cycles
Storage anchor	Horizon table	counted	1 integer/output channel
Storage anchor	Control/storage inventory	counted	80,032 bits; 81,153.208 square micrometers